TrES-5b

R-1645

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_200810` · created 2026-08-02T07:25:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459071.791 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.042
Transit depth (Rp/Rs)^2	2.67 +/- 1.38 [%]
Orbital Inclination (inc)	84.29 +/- 2.37
Airmass coefficient 1 (a1)	0.96 +/- 0.03
Airmass coefficient 2 (a2)	0.03 +/- 0.14
Scatter in the residuals of the lightcurve fit is	3.18 %
Best Comparison Star	#2 - [509, 453]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.042 vs 0.142 (deviation 0.5σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.23σ)
Mid-time O–C	-2.8 min
Depth SNR	3.9
Residual scatter	3.18 %

Light curve

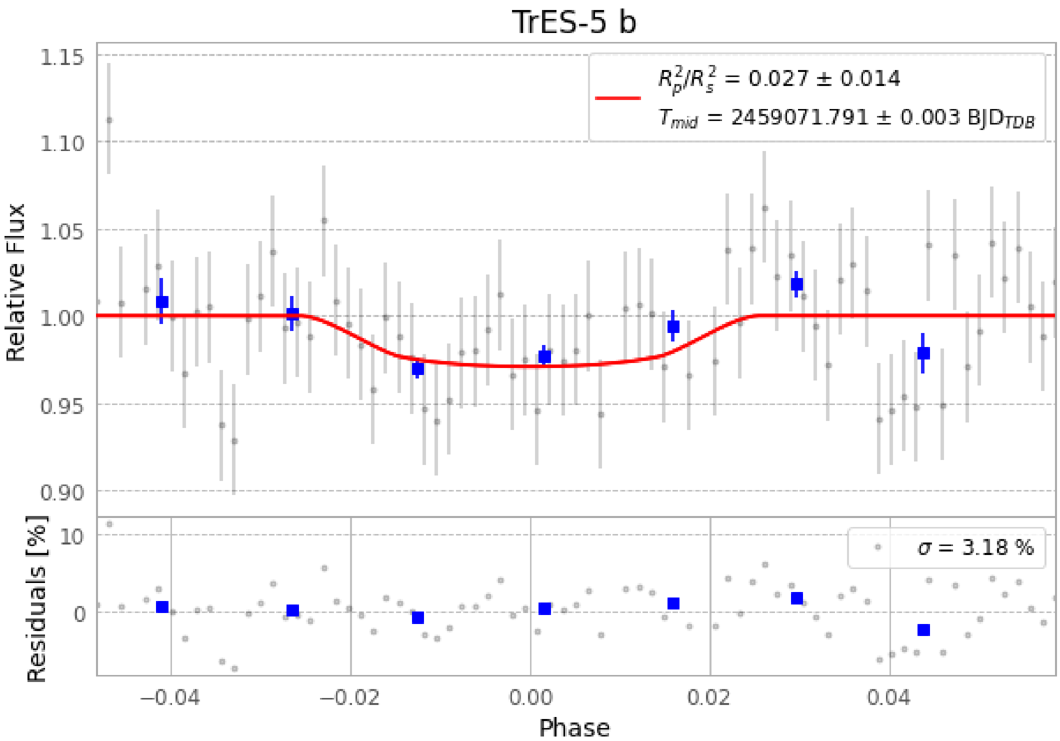

Figure 1 — FinalLightCurve_TrES-5 b_2020-08-09.png (SHA-256 b6ad1c44b36592c1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [494, 127], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1421b472dcf6cff5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `TRES-5200810051214.FITS` → `TRES-5200810090012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-200810.log (7c66d252c5ed…), FinalLightCurve_TrES-5 b_2020-08-09.png (b6ad1c44b365…), FinalLightCurve_TrES-5 b_2020-08-09.csv (c15744c0710f…)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)